

PIATRA NEAMT, Romania

WMO: 151090

Lat: 46.9336N

Lon: 26.3894E

Elev: 1186

StdP: 14.08

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	12.0	3.1	6.7	13.1	6.7	8.1	14.7	26.3	27.4	20.1	34.6	3.8	320	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.1	87.2	67.8	84.4	66.9	81.8	66.1	70.6	82.3	69.1	80.5	67.8	78.7	6.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	103.4	75.1	65.4	98.0	73.5	64.0	93.1	71.9	35.2	82.6	34.0	80.7	32.8	79.0	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	15.2	13.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	27.5	33.2	39.9	49.5	60.2	67.5	69.0	69.5	61.5	50.7	41.3	32.5
	DBStd	16.56	8.83	9.45	8.85	8.59	6.25	5.19	5.09	5.71	7.77	7.93	8.00	9.92
	HDD50	2538	699	471	325	116	7	0	0	0	5	90	282	543
	HDD65	5864	1164	889	779	465	170	34	20	26	153	445	712	1007
	CDD50	2640	0	2	11	101	323	524	589	606	351	111	21	1
	CDD65	490	0	0	0	1	21	107	145	166	49	1	0	0
	CDH74	3740	0	0	0	19	188	797	1017	1311	395	13	0	0
	CDH80	1009	0	0	0	1	29	174	284	428	93	0	0	0
	WSAvg	5.6	5.5	5.6	6.0	6.9	6.1	5.5	5.2	5.4	5.1	5.2	4.9	5.9
	PrecAvg	24.1	0.8	0.8	1.2	2.1	2.9	4.0	4.0	2.4	2.0	1.8	1.1	1.1
(15) Precipitation	PrecMax	36.8	1.7	1.6	2.5	4.2	5.7	10.1	10.7	6.1	4.7	6.9	2.2	2.3
	PrecMin	15.8	0.1	0.3	0.4	0.4	0.7	1.4	0.9	0.7	0.4	0.1	0.0	0.1
	PrecStd	5.1	0.4	0.4	0.5	1.0	1.5	2.1	2.2	1.3	1.1	1.3	0.6	0.7
	PrecStd	5.1	0.4	0.4	0.5	1.0	1.5	2.1	2.2	1.3	1.1	1.3	0.6	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	52.0	59.6	68.3	76.2	83.3	87.2	90.1	91.5	86.3	75.4	66.2	57.4	
	0.4% MCWB	44.2	49.0	51.6	58.3	62.4	70.0	68.8	69.4	65.4	59.8	54.0	47.6	
	2% DB	47.4	54.3	61.2	72.5	78.9	84.2	86.3	87.5	82.1	70.2	60.4	52.1	
	2% MCWB	40.5	45.5	48.7	57.3	61.8	67.8	68.2	67.1	64.1	57.9	51.7	43.9	
	5% DB	43.0	50.0	56.3	68.1	75.7	81.7	82.9	84.7	78.3	66.2	56.5	48.8	
	5% MCWB	37.5	43.2	46.3	54.6	61.2	66.7	67.0	66.8	62.8	55.8	50.1	42.2	
	10% DB	39.7	45.0	52.3	63.4	72.1	79.0	79.4	81.5	74.3	61.7	51.6	45.0	
	10% MCWB	35.5	40.3	43.8	52.8	59.4	65.8	66.4	66.1	61.4	54.1	47.0	39.9	
	0.4% WB	45.0	49.4	53.3	60.6	66.6	74.4	72.4	72.9	68.4	62.5	55.3	47.9	
	0.4% MCDB	51.8	58.0	66.0	73.7	79.3	84.0	85.3	86.0	83.2	70.8	64.9	55.7	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	41.4	46.2	49.8	58.3	64.4	70.2	69.9	70.4	66.1	59.7	52.5	45.0	
	2% MCDB	46.3	53.1	59.5	70.6	74.9	81.0	81.5	82.9	77.3	67.7	59.1	51.5	
	5% WB	38.4	43.5	47.2	55.8	62.5	68.5	68.7	68.7	64.0	57.4	50.4	42.6	
	5% MCDB	42.2	48.5	55.8	66.3	71.8	79.2	79.6	80.4	75.0	64.4	55.6	48.6	
	10% WB	35.9	40.9	44.7	53.7	60.8	66.7	67.4	67.2	62.5	55.2	48.0	40.3	
	10% MCDB	39.6	44.5	50.9	62.4	69.5	76.8	77.4	78.3	72.3	60.5	51.7	45.0	
(35) Mean Daily Temperature Range	MDBR	11.2	12.6	14.9	17.1	18.0	18.2	18.2	19.1	17.6	15.0	11.4	10.5	
	5% DB	17.2	19.2	22.7	25.0	23.0	22.3	22.7	23.3	23.7	21.7	18.5	16.1	
	MCWBR	11.6	11.9	13.2	13.0	10.1	9.4	9.0	8.6	9.3	11.5	10.7	10.1	
	5% WB	15.3	16.2	21.8	23.1	19.8	20.6	19.9	20.9	21.5	19.2	16.7	15.1	
	MCWBR	10.6	10.8	12.9	12.5	9.8	9.8	9.0	8.8	9.2	10.5	10.1	10.0	
	MCWBR	10.6	10.8	12.9	12.5	9.8	9.8	9.0	8.8	9.2	10.5	10.1	10.0	
(40) Clear-Sky Solar Irradiance	taub	0.314	0.345	0.387	0.452	0.425	0.422	0.448	0.457	0.410	0.371	0.340	0.313	
	taud	2.325	2.251	2.201	2.060	2.215	2.261	2.196	2.162	2.255	2.348	2.377	2.354	
	Ebn at Noon	241	256	262	254	267	268	259	250	252	245	230	227	
	Edh at Noon	24	32	39	49	43	42	44	44	36	29	23	21	
(44) All-Sky Solar Radiation	RadAvg	393	637	1014	1392	1772	1865	1875	1702	1204	777	410	308	
	RadStd	53	83	106	125	152	163	151	131	133	94	39	36	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	+1.17	N/A	N/A	N/A	+1.40	N/A	N/A	-261	-364	+144	+32

Nomenclature: See separate page